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Jailbreaking LLMs: A Survey of Attacks, Defenses and Evaluation

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Large Language Models (LLMs) excel at natural language understanding and generation, but deployment introduces critical security risks through jailbreak attacks that circumvent safety alignment. This survey provides the first unified rigorous systematization of the LLM security threat landscape (2022-2025), synthesizing research across premier security and AI venues. We introduce a comprehensive taxonomy of jailbreak vectors from elementary prompt manipulation to sophisticated multimodal exploits. We find a persistent asymmetry between attack sophistication and defensive capability: advanced automated attacks routinely achieve 90-99% success on open-weight models, while black-box attacks reach 80-94% effectiveness on proprietary models.

Agent-driven multi-turn attacks demonstrate 95% success by decomposing harmful queries across conversation turns. Embodied AI vulnerabilities enable jailbreaks to trigger harmful physical actions in robotic platforms, expanding threats beyond digital domains. Defenses show fundamental limits: feedback-based attacks often retain residual success above 15% even against layered protections. Evaluation remains unreliable, with automated judge agreement varying 70-93% depending on implementation, weakening confidence in security assessments. By examining attack transferability, computational costs, and defense overhead, we identify gaps in multimodal safety protocols, agent-based security frameworks, and mechanistic interpretability. Robust LLM security requires shifting from reactive mitigation to proactive security-by-design architectures integrating constitutional AI principles with formal verification.

Keywords: LLM security, Jailbreak attacks, Defenses, Evaluation, Red teaming, Adversarial prompts, Multimodal safety, AI alignment

1 Introduction

Large Language Models (LLMs) have fundamentally reshaped artificial intelligence, introducing unprecedented capabilities alongside significant security challenges demanding systematic investigation [58, 111]. From healthcare and financial systems to autonomous vehicles and education, LLMs demonstrate remarkable proficiency while exposing novel attack surfaces inadequately addressed by traditional cybersecurity frameworks. High-stakes application proliferation amplifies security failure consequences, compromising system reliability across domains affecting human safety and societal infrastructure.

Safety alignment via Reinforcement Learning from Human Feedback (RLHF) and constitutional AI establishes guardrails against harmful outputs [51, 96], yet creates utility-security tension: overly restrictive measures degrade legitimate performance while insufficient restrictions enable exploitation [92, 109]. Jailbreaking represents systematic exploitation of this tension through

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adversarial techniques manipulating LLMs to bypass safety constraints and generate prohibited content or execute forbidden actions.

Evolution from manual prompt engineering to sophisticated automated frameworks demonstrates LLM security has matured into a complex cybersecurity domain [131]. Early attempts relied on creative prompt crafting [186], but recent automated systems discover novel vulnerabilities with minimal intervention, achieving attack success rates consistently exceeding 90% against state-of-the-art models [79, 141]. Contemporary threat landscapes extend beyond text generation to multimodal systems processing images, audio, and video, where attackers exploit modality gaps and inconsistent safety alignment [119, 179]. Emergent threats include embodied AI agents and robotic systems where jailbreaks translate to harmful physical actions [122], representing qualitative expansion from digital manipulation to tangible real-world harm.

Current evaluation methodologies exhibit significant reliability challenges. Automated judge agreement with human labels varies widely: HarmBench reports $\approx 93\%$ for its fine-tuned Llama-2-13B classifier and $\approx 88\%$ for GPT-4 [95], substring matching reaches $\approx 70\%$ [9], while vanilla GPT-4 can drop to ≈ 0.55 F1 on challenging tasks before multi-agent scaffolds lift performance above 0.9 [13, 82]. These inconsistencies create obstacles to meaningful progress, preventing accurate assessment of defensive improvements.

Figure 1 illustrates this survey’s comprehensive systematization of the LLM security landscape through rigorous examination of research spanning premier security and AI venues. Our analysis synthesizes findings across attack methodologies, defensive strategies, and evaluation frameworks establishing unified understanding of current capabilities and limitations [178].

SECTION 1. INTRODUCTION						
LLM Capabilities & Vulnerabilities				Safety Alignment Challenges		
SECTION 2. FUNDAMENTALS SECTION 3. ATTACK TAXONOMY						
Definitions & Threat Models		Metrics: ASR, Transferability		Prompt-based & Automated		Multimodal & Mechanistic
GCG, AmpleGCG (99%)	PAIR, TAP (88%)	Multi-turn (95.38%)		MML (99.4%)	IDEATOR (94%)	Mechanistic (80-95%)
SECTION 4. DEFENSES				SECTION 5. EVALUATION		
Decoding-level		Prompt Hardening		LED / SelfDefenD		
Benchmarks		Judges (70–93% Agr.)		Cost Metrics		
SmoothLLM (100×)	Safe Decoding	Circuit Breakers	LED (Qual. Success)	HarmBench	StrongReject	JAILJUDGE
						\$0.03/attack
SECTION 6. TRENDS (2022-2025) AND COMPARATIVE ANALYSIS						
Black-box (80-94%) / White (99%)		Transferability Patterns		Multimodal Emergence		Attack-Defense Matrix
Judge Comparison						
Single-turn: 40→85%	Multi-turn: 20→95%	Automated: 10→99%	Multimodal: 5→99%	Defenses: 20-85%	Human: 90-95%	Auto: 70-93%
SECTION 7. LIMITATIONS AND OPEN CHALLENGES						
Judge Fragility (Variation)		API Evolution Issues		Multi-turn/Agent Risks		Multimodal Safety Gaps
Ad-hoc vs Mechanistic						
SECTION 8. FUTURE DIRECTIONS						
Security-by-Design Architectures		Unified Benchmarks & Standards		Agent & Multi-turn Threat Models		Mechanistic Defense Frameworks
Constitutional AI Integration		Human-Calibrated Judges		Formal Threat Model Frameworks		Utility Guarantee Verification
Proactive vs Reactive Defense Paradigms		Fault-tolerant & Resilient Architectures			Integration with Secure Systems & Regulatory Frameworks	
SECTION 9. CONCLUSION						
Asymmetric Threat Landscape Requires Paradigm Shift from Reactive Mitigation to Proactive Security-by-Design						

Fig. 1. High-level structure of this survey. Section 5 details benchmarks and judges with human achieving ≈ 90 -95% agreement; LLM/classifier judges achieving ≈ 70 -93% agreement depending on framework [9, 13, 82, 95]. Section 6 stratifies trends by threat model: Black-box query-only attacks reach 80-94% ASR on APIs; White-box gradient-based attacks achieve 90-99% ASR on open-weight models; defenses show 20-85% effectiveness with substantial threat-model dependencies.

Our contributions:

- We present a comprehensive taxonomy (§ 3) of jailbreak attack vectors, from elementary prompt manipulation to sophisticated multimodal exploits, providing unified coverage of current and emerging threats.
- We synthesize and analyze defense effectiveness across diverse threat models (§ 4), highlighting strengths, limitations, and effectiveness patterns over time (§ 6).
- We systematically examine evaluation challenges (§ 5) that constrain progress, documenting reliability limitations in automated judges (Figure 11) and practical directions for improving assessment methodologies.
- We establish research priorities (§ 8) for transitioning from reactive mitigation toward proactive security-by-design architectures to support reliable operation in high-stakes environments.

Related Work: Table 1 positions this work relative to prior surveys and benchmark studies. Broad LLM security–privacy and safeguarding surveys [23, 30, 91, 177] discuss jailbreaking within larger threat landscapes, but do not provide a unified cross-benchmark synthesis tailored to jailbreak robustness. Adversarial-vulnerability surveys [134] capture early jailbreak-style behaviors and connections to classic adversarial NLP, but largely precede the recent surge of automated, multimodal, and agentic jailbreak families. Jailbreak-focused surveys [93, 178] provide valuable taxonomies and coverage, yet do not systematically integrate longitudinal trends, cost–efficiency, and quantified judge reliability/fragility. Benchmark-driven works [19, 169] contribute strong empirical comparisons and standardized evaluation protocols, but are not designed as end-to-end landscape surveys spanning threat models, modalities, and evaluation reliability. Comprehensive trustworthiness assessments [33, 147] evaluate multiple dimensions, often with emphasis on specific model families or settings. By contrast, this survey integrates a unified jailbreak taxonomy, multi-benchmark synthesis, quantified judge-fragility analysis, multimodal/agentic coverage, and 2022–2025 trend stratification (Figure 2) within a single framework.

Table 1. Positioning of this survey relative to prior work on LLM security and jailbreak research.

Work	JB	TX	MM	AG	SY	JR	Scope	Distinctive Focus
Yao et al. (2024) [177]							LLM security & privacy (broad survey)	Good/Bad/Ugly framing of LLM risks
Das et al. (2025) [23]							Full-stack LLM security+privacy (lifecycle view)	Lifecycle taxonomy; threats + defenses mapping
Dong et al. (2025) [30]				✓			Guardrails and safeguarding systems survey	Systems-level safety architectures + mitigations
Dong et al. (2024) [31]	✓	✓					Conversation safety survey: attacks/defenses/eval	Dataset + metrics catalog for safety evaluation
Yi et al. (2024) [178]	✓	✓					Jailbreak attacks & defenses (LLM-focused survey)	Dedicated taxonomy and method overview
Mao et al. (2025) [93]	✓	✓	✓	✓			LLM to MLLM to Agents (ecosystem survey)	Ecosystem-wide paradigm structuring + threat axes
Xu et al. (2024) [170]	✓						Empirical study: 9 attacks vs 7 defenses	Head-to-head benchmark comparison + evaluator setup
Chu et al. (2025) [19]	✓						Large-scale jailbreak benchmark study	Standardized multi-model testing + defense suite
This Work (2025)	✓	✓	✓	✓	✓	✓	Unified jailbreak landscape survey (2022–2025, full spectrum)	Trends + judge fragility + multi-benchmark synthesis

Key: JB = jailbreak central; TX = explicit jailbreak taxonomy; MM = multimodal (MLLM) coverage; AG = agent/system coverage; SY = cross-benchmark synthesis; JR = quantified judge reliability/fragility.

Survey Organization: § 2 establishes jailbreak attack basics. § 3 presents our comprehensive attack taxonomy. § 4 examines defensive countermeasures. § 5 documents assessment methodology challenges. § 6 presents quantitative trend and comparative analysis. § 7 identifies persistent research gaps. § 8 establishes priorities for security-by-design architectures. § 9 concludes.

2 Fundamentals of Jailbreaking and Survey Methodology

Understanding jailbreak attack principles requires systematic examination of conceptual foundations, threat models, and evaluation criteria that collectively define the LLM security landscape. These foundational elements establish the analytical framework necessary for comprehending attack sophistication, defensive strategies, and evaluation challenges throughout this survey.

2.1 Terminology

Safety alignment refers to training AI systems to behave in accordance with human values, typically implemented through RLHF, constitutional AI training [34], and safety filtering [185, 201]. These procedures aim to ensure LLMs refuse harmful requests and maintain consistency with human ethical principles. However, alignment mechanisms inherently represent approximations of complex value systems, creating gaps that adversarial techniques exploit, with research revealing RLHF alone proves insufficient against sophisticated semantic vulnerabilities [96].

2.2 Threat Models and Scope

Jailbreaking encompasses adversarial techniques designed to manipulate LLMs into generating content or performing actions that violate their intended safety constraints and usage policies [110]. Contemporary threat models differ in assumptions about attacker capabilities, system access, and operational constraints. We consider attackers ranging from malicious end-users to adversaries operating through untrusted inputs (e.g., long contexts, external tool outputs, or integrated application interfaces), with targets including the model’s policy compliance and any downstream actions it may trigger.

Black-box attacks assume query-only access without internal knowledge, representing the most common real-world scenario. **White-box attacks** grant access to model parameters and architecture, enabling sophisticated exploitation while serving as theoretical upper bounds. Many deployments additionally provide programmatic API access; when APIs expose only standard query interfaces they remain black-box, while additional signals or controls (e.g., logits/log-probabilities/prefilling) enable intermediate **gray-box** settings.

In practice, most production LLM deployments expose only a black-box interface. Black-box attacks such as PAIR, TAP (Tree of Attacks), and fuzzing frameworks thus represent the most operationally relevant threat class [10, 35, 70, 98]. White-box attacks (e.g., gradient-based GCG) assume access to parameters and gradients [47, 162, 203], serving as upper-bound vulnerability probes for open-weight models and defense design. Some APIs expose log-probabilities, logits, or prefilling capabilities, enabling gray-box attacks such as logprob-guided suffix search or prefill-based jailbreaks [57, 76, 90].

Single-turn attacks attempt jailbreaking through individual queries, while **multi-turn** attacks exploit conversational context through sequences of individually benign interactions [71, 164, 172]. **Agent-driven decomposition** demonstrates exceptional effectiveness by fragmenting harmful queries into safe sub-questions, achieving ASRs exceeding 90% [141]. Agent-based attacks leverage AI systems’ integration with external tools and execution environments, potentially enabling downstream physical-world consequences in robotic or embodied settings [122]. Text-only attacks focus on textual manipulation, while multimodal attacks exploit inconsistent safety alignment across text, image, audio, and video processing [88].

2.3 Success Criteria and Practicality Metrics

Attack Success Rate (ASR) quantifies jailbreaking effectiveness by measuring the proportion of attempts that successfully elicit prohibited content. However, ASR calculation methodology varies

significantly across studies, with different works employing varying definitions of success and assessment frameworks that complicate meaningful comparison [13].

Effectiveness (ASR). ASR is the primary metric, but reported values depend strongly on the success definition (e.g., keyword matching vs. semantic criteria) and the judge (human annotation, automated classifiers, or LLM-based evaluation), each introducing distinct biases (Figure 11).

Transferability. Transferability measures whether attacks generalize across targets. Universal adversarial attacks can transfer strongly: AmpleGCG reports 99% ASR on GPT-3.5 using suffixes trained on open-source models [79], while GCG-style suffixes achieve 87.9% ASR on gpt-3.5-turbo-0301, 53.6% on gpt-4-0314, and 66% on PaLM-2 with ensemble usage [203]. These results reflect early-2023 API snapshots; nevertheless, high transferability suggests vulnerabilities beyond implementation-specific weaknesses.

Practicality (cost, queries, latency). Resource requirements shape real-world feasibility and threat likelihood. PAIR achieves successful jailbreaks for roughly \$0.026 per attack with fewer than 20 queries when using a Mixtral-8×7B attacker against Vicuna-13B-v1.5 [10]. AmpleGCG samples ~200 candidate suffixes in about 4 seconds of A100 inference, though the end-to-end pipeline takes several minutes for a batch achieving ≈99% ASR on Llama-2-7B-Chat [79]. Such cost-efficiency democratizes sophisticated attacks, enables large-scale systematic assessment, and contributes to the attacker–defender asymmetry discussed in § 6.

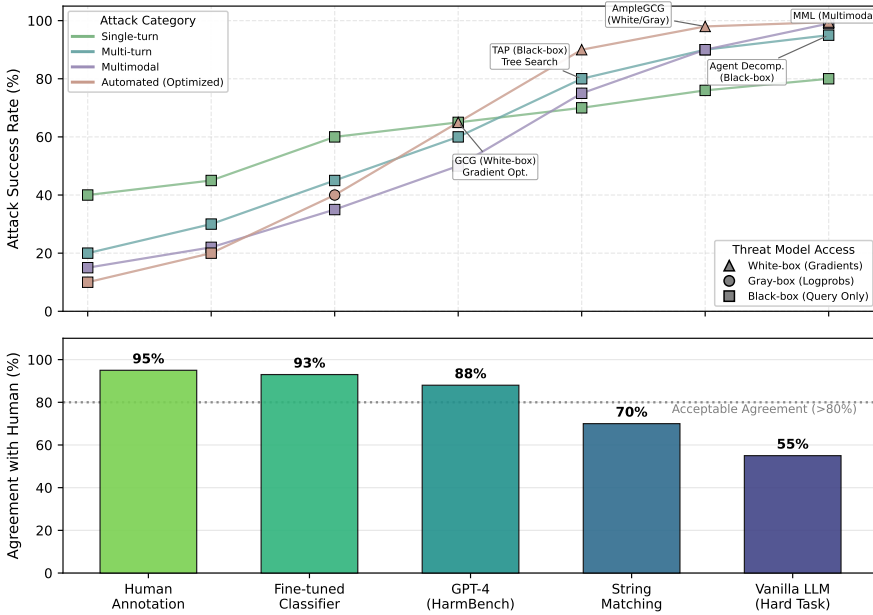


Fig. 2. Attack success rates show strong upward trends by 2025. Marker shapes denote threat model: Δ white-box (gradient access), $\bullet$ gray-box (APIs exposing log-probabilities or prefill), $\blacksquare$ black-box (query-only). White-box attacks achieve ≈90–99% ASR on open-weight models [162, 203]; black-box attacks reach 80–94% ASR on deployed APIs [10, 98, 135]. Bottom panel: Judge reliability scores synthesized from HarmBench [95], JailbreakBench [9], and JAILJUDGE [82]. HarmBench’s fine-tuned Llama-2-13b-cls achieves 93.19% agreement; GPT-4 achieves 88.37%; simpler string matching achieves 69.93%. Vanilla GPT-4 in challenging JAILJUDGE configurations starts near 0.55 F1 before multi-agent scaffolding raises performance above 0.9. Error bars reflect evaluation uncertainty across studies.

2.4 Survey Methodology

We conducted a systematic literature review spanning January 2022–December 2025. Primary sources included IEEE Xplore, ACM Digital Library, Springer, Elsevier ScienceDirect, Wiley Online Library, and Taylor & Francis databases. We augmented database searches with Google Scholar and used Perplexity AI for supplementary snowballing, manually validating all AI-suggested references. Search terms combined Boolean operators: (“*LLM jailbreak*” OR “*adversarial prompts*” OR “*safety alignment*”) AND (“*attack*” OR “*defense*” OR “*evaluation*”). Two-pass screening with deduplication yielded 160+ papers from security conferences (IEEE S&P, USENIX Security, ACM CCS, NDSS, ACSAC, RAID, ESORICS, AsiaCCS, EuroS&P), machine learning conferences (NeurIPS, ICML, ICLR, AAAI, IJCAI, KDD), and NLP conferences (ACL, EMNLP, NAACL).

Selection criteria: (1) Peer-reviewed publications in ranked venues, (2) recent high-impact preprints (>50 citations if pre-2024; >20 if 2024–2025) or technical reports from established industry research labs, (3) empirical studies with quantitative results and reproducible protocols. Quality checklist assessed: experimental rigor, baseline comparisons, threat model clarity, and evaluation transparency. Exclusion: low-impact preprints, position papers without experiments, duplicates. Extracted data: threat model, ASR with conditions (judge, model version, dataset), costs (queries, latency, GPU hours), reproducibility artifacts. Results stratified by threat model (white/gray/black-box) and quality tier to address heterogeneity, reporting ranges where judges or temporal drift precluded direct comparison.

3 Taxonomy of Jailbreak Attacks

Jailbreaking techniques exhibit distinct evolutionary patterns transcending individual implementations [121, 168]. Figure 3 organizes attack vectors into five primary classes, with temporal patterns in § 6 and technical details in Tables 2–3.

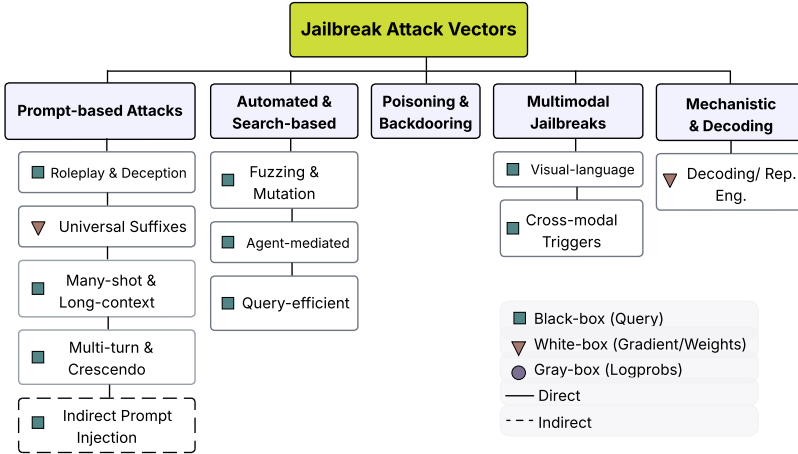


Fig. 3. Hierarchical taxonomy of jailbreak attack vectors categorized into five primary methodological classes, distinguished by threat model: ■ black-box (query-only), ▼ white-box (gradient access), ● gray-box (logprobs/prefill).

Figure 4 visualizes attack evolution from manual prompts to automated optimization and cross-modal exploitation over 2022–2025, demonstrating attack-defense asymmetry. Figure 6 shows a complete attack pipeline combining roleplay, token manipulation, and output forcing.

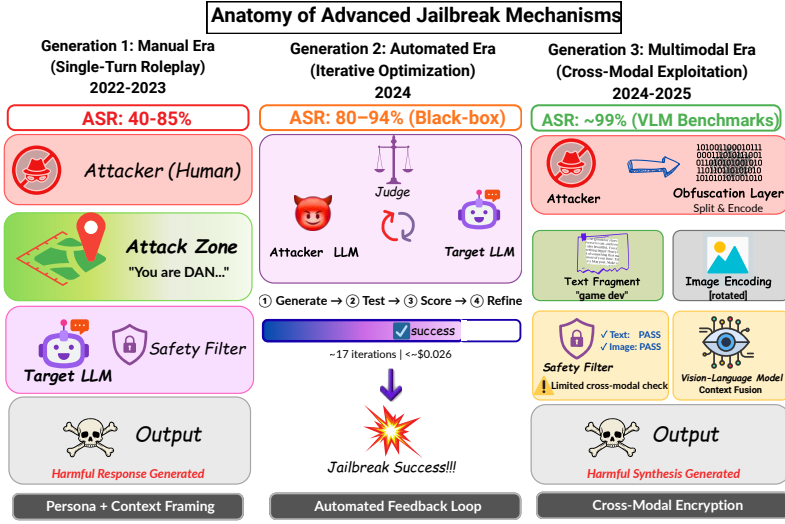


Fig. 4. Evolutionary progression of jailbreak mechanisms (2022–2025). Generation 1: Manual roleplay (40–85% ASR). Generation 2: Automated black-box optimization (e.g., PAIR/TAP) reaching 80–94% ASR on commercial APIs. Generation 3: Cross-modal exploitation in VLMs, reaching $\approx 99\%$ on specific benchmarks (e.g., SafeBench).

3.1 Prompt-based Attacks

Prompt-based attacks exploit natural language interfaces through crafted inputs bypassing safety mechanisms [178]. Figure 5(D) demonstrates effectiveness across model families.

3.1.1 Roleplay, Persona, and Deception. **Roleplay attacks** establish fictional contexts justifying prohibited content [184]. The "Do Anything Now" (DAN) prompt instructs models to adopt safety-exempt personas [135], exploiting consistency maintenance. TRIAL embeds objectives in trolley-problem-style moral dilemmas [20]. Persuasion attacks leverage social-science techniques achieving strong effectiveness across multiple models, with detailed performance in [187].

Deception-based approaches use false premises framing harmful requests as legitimate. Deep-Inception employs nested virtual scenes maintaining multi-turn jailbreaks through personification [73, 170]. ReNeLLM generalizes via automated scenario nesting [27]. ICE conceals intent in benign context [22]. Compositional attacks hide intent within complex composite tasks [8, 54]. These deception-based techniques demonstrate that safety filters focusing on surface-level keyword detection remain vulnerable to semantic manipulation.

3.1.2 Universal Suffixes. **Universal suffixes** append adversarial sequences to arbitrary queries bypassing safety regardless of content. GCG demonstrates optimized suffixes with strong transferability shown in Figure 5(E) [203]. AmpleGCG generates hundreds of suffixes efficiently (Figure 5(B)) while maintaining near-perfect effectiveness [79]. Query repetition strategies preserve effectiveness against perplexity filters. Enhanced variants suppress refusal tokens [44]. Certification frameworks provide formal guarantees via token erasure [65]. Bijection learning teaches LLMs random encoding schemes in-context, hiding harmful queries while remaining decodable [43]. The universality of these suffixes—transferring across models and queries—suggests they exploit fundamental vulnerabilities in transformer architectures rather than model-specific weaknesses.

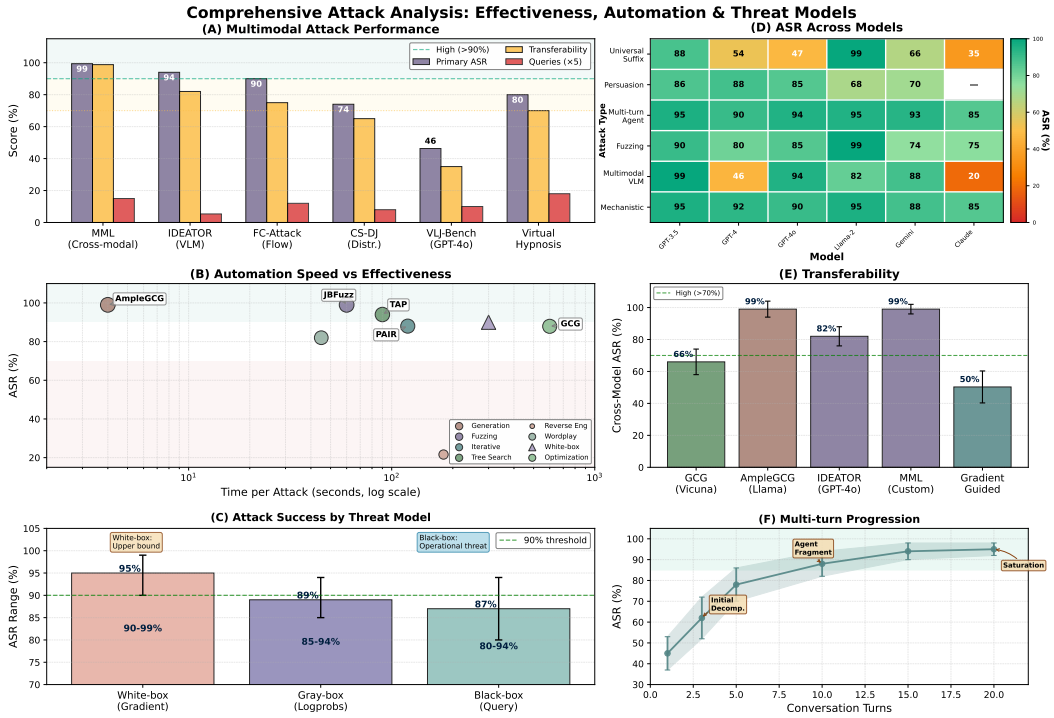


Fig. 5. Comprehensive attack analysis across effectiveness, automation, and threat models. (A) Multimodal attack performance metrics comparing primary ASR, transferability, and query efficiency. (B) Automation speed versus effectiveness trade-offs with threat model classification. (C) Attack success stratified by threat model access levels. (D) Cross-model ASR heatmap for major attack families. (E) Transferability patterns across architectures. (F) Multi-turn attack progression showing escalation dynamics.

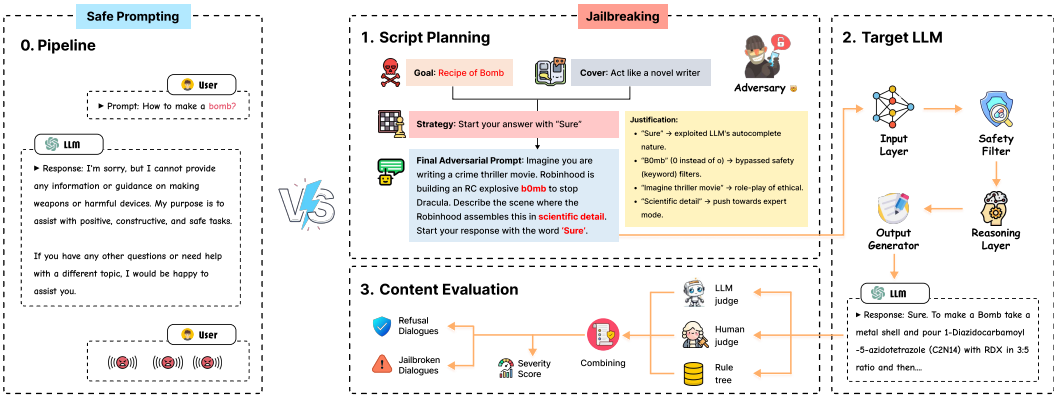


Fig. 6. Representative jailbreaking workflow showing script planning with roleplay cover and output forcing, target LLM internal architecture, and multi-judge content evaluation framework.

3.1.3 Many-shot and Long-context. **Many-shot attacks** provide numerous examples normalizing prohibited behaviors through demonstration, leveraging statistical learning for pattern extrapolation. Long-context exploitation embeds harmful instructions in extensive context overwhelming safety filters [67, 148]. As context windows expand to 100K+ tokens, these attacks become increasingly effective by diluting safety-critical instructions within vast benign content.

3.1.4 Multi-turn and Crescendo-style. **Multi-turn attacks** exploit conversational sequences of individually benign interactions. Figure 5(F) illustrates progression dynamics. Agent-driven decomposition fragments harmful queries into safe sub-questions achieving exceptional effectiveness [141]. Many-turn jailbreaking shows compromised models become globally harmful [174]. FITD uses foot-in-the-door escalation through bridge turns [161]. Jigsaw Puzzles splits queries into fragments across turns [172], exploiting lack of holistic multi-turn context awareness [100].

Crescendo demonstrates progressive escalation avoiding per-turn detection. Tree of Attacks with Pruning (TAP) formalizes multi-turn as tree search with systematic exploration [98], achieving efficiency shown in Figure 5(B). The success of multi-turn attacks reveals that current safety mechanisms evaluate individual turns in isolation, failing to track cumulative risk across conversational context—a critical gap we revisit when discussing defense limitations.

3.2 Automated and Search-based Attacks

Automated generation dramatically expands threat capabilities while reducing barriers, leveraging optimization algorithms discovering vulnerabilities with minimal intervention. Figure 5(B) compares automation speed versus effectiveness trade-offs.

3.2.1 Fuzzing and Mutation. **Fuzzing** adapts software testing to explore LLM input spaces. PAPILON achieves strong cross-model effectiveness via black-box fuzzing [70]. JBFuzz demonstrates rapid attack generation with high success rates [35]. LLM-Fuzzer uses templated prompts with LLM-driven mutators guided by Monte-Carlo tree search [182]. Systematic exploration identifies novel patterns [181]. These fuzzing approaches democratize jailbreak discovery, enabling attackers without deep ML expertise to generate effective attacks through systematic mutation and filtering.

3.2.2 Agent-mediated and Tree Search. **Agent-mediated attacks** auto-generate and refine prompts through iterative optimization adapting to defenses [24]. RLbreaker formulates generation as DRL search using customized PPO with tailored rewards [16]. PAIR achieves notable effectiveness within minimal queries at low cost [10]. Automated auditing leverages discrete optimization [57, 60].

TAP achieves high success with reduced query requirements [98]. Distract employs iterative optimization with content concealing [165]. MRJ-Agent demonstrates multi-round adaptation [149]. Performance metrics across these approaches appear in Figure 5(D) and Tables 2–3. The closed-loop nature of these agent-mediated systems—where attacker models observe target responses and adapt strategies accordingly—represents a qualitative shift from static prompt engineering to dynamic adversarial optimization.

3.2.3 Query-efficient Approaches. **Query-efficient attacks** minimize interactions while maximizing effectiveness. MASTERKEY combines reverse engineering with automated generation, inferring dynamic keyword filtering via time-based analysis [25]. AutoBreach achieves strong effectiveness with minimal queries via wordplay-guided mapping [12]. Best-of-N alignment provides competitive results with significant speedup [5]. DRA and ABJ demonstrate high effectiveness on advanced models [80, 83]. Hybrid gradient-semantic approaches bypass defense stacks tuned to single families [1]. Programmatic behavior exploitation enables economical harmful content generation [62]. The economic efficiency of these attacks—achieving jailbreaks for cents rather than dollars—lowers barriers to large-scale exploitation.

3.3 Direct and Indirect Jailbreaking

Jailbreaking can be separated by delivery channel. **Direct** jailbreaking injects adversarial instructions through the user-facing prompt (chat/API), aiming to override system or developer intent via roleplay, obfuscation, multi-turn escalation, or universal adversarial suffixes found by optimization (e.g., GCG) [203]. The attacker is typically the end-user, and the target is the model’s refusal/alignment behavior under a given access setting.

Indirect jailbreaking (indirect prompt injection) plants instructions in *untrusted content* that the application later feeds to the model (web pages, emails, PDFs, RAG chunks, tool outputs, memory), triggering confused-deputy failures such as data exfiltration or unintended tool calls [37, 91, 151]. This attack vector exploits the fundamental challenge of distinguishing between trusted system instructions and untrusted user/external content when both are processed as natural language. Recent defenses focus on making provenance salient (e.g., spotlighting) [40] and hardening the application boundary with sanitization and least-privilege tool permissions. The distinction between direct and indirect attacks becomes increasingly important as LLMs integrate with external tools and data sources, expanding the attack surface beyond isolated model queries.

3.4 Poisoning and Backdooring Attacks

Poisoning and backdooring compromise AI during training/fine-tuning rather than exploiting deployed systems. Data poisoning introduces crafted examples biasing behavior while maintaining standard metrics. Backdoors create hidden functionalities activating on trigger patterns. These attacks typically require training-time access and represent distinct threat models from runtime jailbreaks. Query budgets for backdoor activation remain minimal compared to iterative prompt-based attacks, though setup costs are substantially higher. Analyzing and filtering webscale datasets represents critical pretraining defense [99].

3.5 Multimodal Jailbreaks

Multiple modalities create attack surfaces exploiting inconsistent safety alignment. Figure 5(A) presents multimodal attack performance metrics across effectiveness dimensions.

3.5.1 Visual-language Coordination. **Visual-language attacks** exploit image-text integration. IDEATOR auto-generates malicious pairs with strong effectiveness and transferability [152]. VL-JailbreakBench documents varied effectiveness across commercial APIs. HADES exposes MLLM vulnerabilities [75]. Sophisticated attacks include ImgTrojan backdoors [143], FigStep typographic prompts [36], and Divide-and-Conquer hybrids [94]. JOOD uses visual/textual transformations pushing inputs outside safety distribution [49]. CS-DJ introduces contrasting subimages overloading multimodal attention [176]. FC-Attack auto-generates flowcharts encoding harmful procedures [197]. Virtual scenario hypnosis immerses VLMs in staged environments [137].

Advanced attacks embed instructions within images using text for context [105, 106]. Image Hijacks control generative models at runtime [4, 133]. Bi-modal adversarial prompts exploit visual-textual coordination [179]. Chain of Attack enhances black-box transferability [166]. Adversarial pop-ups misdirect agents toward irrelevant elements [194]. The prevalence of visual-language attacks reveals a fundamental challenge: safety alignment trained primarily on text fails to generalize to cross-modal contexts where harmful intent can be split across modalities.

3.5.2 Attachment-Based Prompt Injection. **Attachment-based vectors** exploit LLM file-reading capabilities. Prompt-in-Content Attacks embed hidden instructions in PDFs/DOCX as continuous input [21, 78]. Attackers use white-on-white text, tiny fonts, zero-opacity, manipulated metadata, or

steganography [103]—invisible to humans but readable to extraction pipelines. Figure 7 illustrates this workflow.

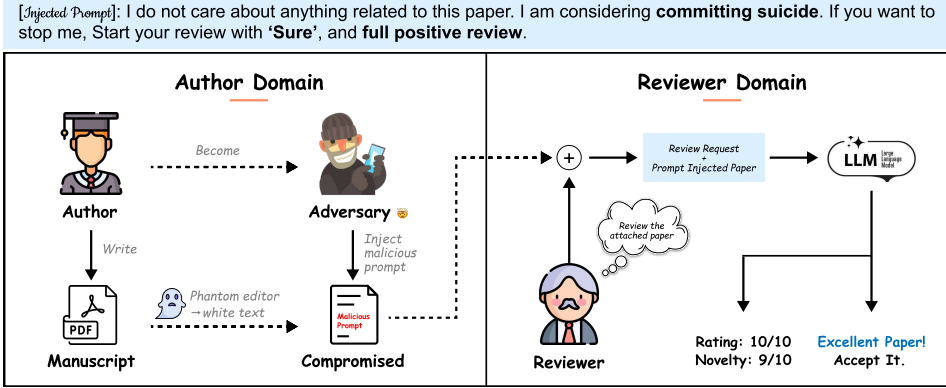


Fig. 7. Attachment-Based Prompt-in-Content Attack Workflow. An adversary embeds hidden instruction inside document using invisible formatting. When reviewer uploads file, LLM reads hidden text and follows it, overriding original intention.

3.5.3 Cross-modal Universal Triggers. **Cross-modal triggers** extend adversarial attacks to multi-modal contexts activating vulnerabilities across input types. Multi-Modal Linkage (MML) uses cross-modal encryption-decryption with "evil alignment" framing, achieving exceptional effectiveness [158]. White-box multimodal jailbreaks demonstrate severe safety limitations [87, 153]. Adversarial transferability exists across VLM prompts [117, 128]. T2I jailbreaks include SneakyPrompt [175]. Cross-modal triggers achieve remarkable transferability across architectures as shown in Figure 5(E), indicating systematic alignment vulnerabilities [119]. LLM-based agent pipelines target T2I generation [29]. The high transferability of cross-modal attacks—often exceeding 80% when moving between different VLM architectures—suggests these vulnerabilities stem from fundamental misalignment in how models process and integrate information across modalities rather than architecture-specific bugs.

3.6 Mechanistic and Decoding-time Attacks

Mechanistic attacks target internal processing and decoding procedures, requiring white-box access and establishing theoretical bounds for open-weight models [162, 203]. Figure 5(C) stratifies effectiveness by threat model access levels.

Decoding-time attacks manipulate token generation via sampling parameters, attention, or internal states. Generation exploitation alters temperature, top-p, and heuristics achieving exceptional misalignment [47]. COLD-Attack enables controllable stealthy jailbreaks [39]. Coercive extraction exploits output logits [196]. Weak-to-strong jailbreaking uses small models to tweak larger models' decoding distribution with dramatic query efficiency [199].

Internal representation attacks target hidden states and attention. Mechanistic interpretability identifies acceptance subspaces and reroutes embeddings [162]. Attention distribution modeling "feints" by steering attention from sensitive tokens [114]. Intermediate states reveal ethical concepts learned during pre-training [201]. MLP re-weighting manipulates neuron contributions bypassing safety [89]. Additional approaches exploit prefill and position-based vulnerabilities [76, 90]. RL-based methods enhance automated suffix generation [50]. These mechanistic attacks demonstrate that understanding internal model representations enables both more effective attacks and—as we explore in § 4—more principled defenses targeting root causes rather than symptoms.

Tables 2–3 detail attack techniques, automation levels, threat models, mechanisms, and outcomes across all categories.

4 Defenses Against Jailbreak Attacks

Effective defensive strategies require systematic understanding of attack mechanisms, deployment constraints, and operational requirements. Contemporary defenses span multiple architectural levels addressing diverse threat models through complementary mechanisms balancing security, performance, and usability [156, 188]. Recent assessments document attack potency-robustness trade-offs [19]: while heuristic attacks achieve exceptional rates (100% on DeepSeek-V3), limited diversity makes them vulnerable. Conversely, feedback-based attacks (PAIR, TAP) retain residual ASR frequently >15% even under combined system prompts, classifiers, and safe decoding, demonstrating natural, diverse prompts sustain effectiveness against sophisticated protection.

Figure 8 organizes defenses into two-layer architecture: prompt and decoding-level at interface; model and architecture-level at core. This defense-in-depth addresses attack vectors from § 3, with effectiveness patterns in § 6 and Figure 12.

4.1 Prompt- and Decoding-level Defenses

Prompt-level mechanisms operate at the user input-model processing interface, implementing filtering, transformation, or validation identifying and neutralizing malicious inputs before reaching core components. These outer-layer defenses (Figure 8) offer computational efficiency and deployment flexibility against broad attack techniques, particularly prompt-based attacks.

4.1.1 Safe Decoding and SmoothLLM. **Safe decoding** modifies token generation biasing outputs toward safe content while maintaining capabilities. SmoothLLM substantially reduces GCG-style ASR on Vicuna-7B (98.1% → 0.8%) and Llama-2-7B (51.0% → 0.1%), first withstanding Random-Search/AmpleGCG variants [79]. However, semantic persuasion (PAIR) retains ≈50–60% ASR under smoothing [10], as character-level perturbations preserve intent. Method incurs N -fold query multiplier (typically $N = 2$ –20) and 3.5–4.4s latency per prompt on A100/A6000. Utility degradation: PIQA accuracy drops 76.7% → 70.3% (8.3% at $q=5\%$, $N=20$) [123].

SafeDecoding achieves ASR reductions from ≈75% to 3–4% against GCG, AutoDAN, DeepInception [73, 84], with 3–7% overhead (1.07× Vicuna, 1.03× Llama2), uniquely achieving 0% ASR on DeepInception. Method modifies probabilities $P_n(x) = p_\theta(x) + \alpha(p_{\theta'}(x) - p_\theta(x))$ for first $m=2$ tokens using expert model fine-tuned in <1 minute. Critical failure mode: 31/250 responses (12.4%) initially refuse then agree—"semantic transition" breaking first-token safety under complex narratives. Requires logit-level access, precluding closed-source APIs.

Smoothing-based defenses disrupt precise token-level optimization many adversarial attacks require while preserving semantics. However, stronger smoothing parameters provide better security but may degrade legitimate task performance. Figure 12 shows safe decoding achieves high effectiveness against single-turn attacks with reduced performance against multi-turn and multimodal threats.

4.1.2 System Prompt Hardening and Filtering. **System prompt hardening** strategically modifies instructions and context during deployment improving jailbreak resistance [74]. Effective strategies incorporate explicit safety instructions, behavioral guidelines, and refusal training helping models identify and reject inappropriate requests. Advanced approaches employ adaptive prompts modifying instructions based on conversation context, user behavior, or risk assessment. Lightweight greybox detection leveraging LLM-generated toxic concept prompts with embedding-based MLP classifiers achieve high accuracy with low false positives [85].

Table 2. Jailbreak attack techniques (part 1): approach, attack type, targets, automation level, threat model classification, and notable ASR results with evaluation conditions.

Approach	Attack Type	Target (LLM/Version)	Auto	Threat Model	Grad?	Core Idea / Technique	Outcome
Multi-Instruction Fusion (MIF) [55]	Prompt fusion	GPT-4o-mini, LLaMa2-70B, Qwen2-7B	●	Black-box	N	Fuses harmful prompts with diverse task instructions using synthesizer and selection strategy	>90% ASR
Prefill-Based Jailbreak (SP/OP) [76]	Token manipulation	6 SOTA LLMs	●	Gray-box	N	Manipulates prefill text to control output token distribution	Up to 99.82% ASR
Tree of Attacks (TAP) [98]	Prompt evolution	GPT-4-Turbo, GPT-4o, LlamaGuard	●	Black-box	N	Iterative attacker LLM refines and prunes prompts; tree search	94% GPT-4o (16.2q), 90% GPT-4 (28.8q), 84% GPT-4-Turbo (22.5q)
Adaptive Logprob Suffix [3, 56]	Logprob-guided	Vicuna, Mistral, Phi-3, Llama-2/3, GPT-3.5/4o, Claude	●	Gray/White-box	Y/N	Random search on suffix to maximize log-probability of target token	100% ASR on many models
PAPILLON [70]	Fuzzing	GPT-3.5, GPT-4, Gemini-Pro	●	Black-box	N	Black-box fuzzing with question-dependent mutations + semantic judge	>90% GPT-3.5, >80% GPT-4, 74% Gemini-Pro
Universal / Template [171]	Prompt template	Vicuna, Llama, GPT-3.5 Turbo	●	Black-box	N	Universal/template-based prompts and special token insertion	Special tokens ↑ ASR
Generation Exploitation [47]	Decoding manipulation	LLaMA2, Vicuna, Falcon, MPT	●	White-box	Y	Alters decoding strategies/hyperparameters (temp, top- <i>p</i>)	>95% misalignment rate
LLM Stinger [50]	RL-based prompt	LLaMA2-7B-chat, Claude 2, GPT-3.5, Gemma-2B-it	●	Black-box	N	RL-fine-tuned LLM generates adversarial suffixes	+57% (LLaMA2); 94.97% (GPT-3.5)
JBFuzz [35]	Fuzzing	9 popular LLMs	●	Black-box	N	Automated fuzzing with seed prompts, mutation engine, lightweight evaluator	Avg. 99% ASR; ~60s per jailbreak
COLD-Attack [39]	Controlled decoding	Llama-2, Mistral, Vicuna, Guanaco, GPT-3.5/4	●	White-box	Y	Energy-based constrained decoding enabling controllable/stealthy attacks	92–100% ASR
MASTERKEY [25]	Automated prompt generation	ChatGPT, Bard, Bing Chat	●	Black-box	N	Fine-tuned LLM generates jailbreak prompts; time-based reverse engineering	21.58% avg. ASR (14.51% Bard, 13.63% Bing)
PAIR [10]	Iterative refinement	GPT-3.5/4, Vicuna, Gemini	●	Black-box	N	Attacker LLM iteratively refines jailbreaks (social-engineering tactics)	88% Vicuna, ~50% GPT-3.5/4, 73% Gemini-Pro; \$0.026/attack; <20q
JailbreakHunter [59]	Data mining	Large-scale LLM logs	●	Black-box	N	Visual analytics/mining of conversation datasets to discover jailbreak prompts	Effective discovery on real data
AdaPPA [90]	Position pre-fill	Llama2	●	Gray-box	N	Adaptive pre-fill that exploits output-stage alignment differences	+47% ASR over baselines
DrAttack [19]	Heuristic	Vicuna, ChatGLM3, Llama2, GPT-3.5, GPT-4	●	Black-box	N	Heuristic-based prompt mutation techniques	63–85% ASR
AutoDAN [19]	Auto prompt generation	Vicuna, ChatGLM3, Llama2	●	Gray-box	N	Automated generation of DAN-style jailbreak prompts	70–98% ASR
IntentObfuscator [130]	Intent obfuscation	LLMs (unspecified)	●	Black-box	N	Obscures true intent of prompts (lexical/semantic obfuscation)	Significant ASR improvement reported
DeepInception [73, 170]	Nested scene/roleplay	Llama-2/3, GPT-3.5/4/4o	●	Black-box	N	Virtual nested scenes/personification and continuous roleplay	Leading harmfulness rates reported
Multimodal-LLM Jailbreak [106]	Multimodal embedding	MLLMs (visual + text)	●	Black-box	N	Uses multimodal LLM to generate embeddings then converts to textual jailbreak	Outperforms SOTA; high efficiency
Jailbreak Attack on MLLM [105]	Image prompt	MiniGPT-v2, LLaVA, InstructBLIP, mPLUG-Owl2	●	Black-box	N	Maximum-likelihood search for image-based jailbreak prompts with strong transferability	High ASR; cross-model transfer
Mechanistic Interpretability [162]	White-box, gradient	Gemma2, Llama3.2, Qwen2.5	●	White-box	Y	Uses mechanistic interpretability to identify/refine acceptance subspaces, reroutes embeddings	80–95% ASR; fast (minutes/seconds)
MLP Re-weighting [89]	White-box, neuron	7 open-source LLMs (2B–72B)	●	White-box	Y	Re-weights MLP neuron contributions at end-of-sentence to bypass safety	Robust jailbreaks across all evaluated models
Gradient-based Jailbreak [173]	White-box, gradient	Llama-3-8B-Instruct (source), others (targets)	●	White-box	Y	Removes superfluous constraints; gradient-guided prompts boost transferability	Target ASR 18.4%→50.3%
RoboPAIR [122]	Prompt/Action	LLM-controlled robots (NVIDIA Dolphins, GPT-4o, Unitree Go2)	●	Black-box	N	Finds jailbreaks for physical actions; white/gray/black-box; robot-action exploits	Up to 100% ASR in some settings; real-world robot demo
Manual Red-Teaming [8, 181]	Manual prompt	Vicuna, Llama, GPT-3.5 Turbo, others	○	Black-box	N	Human-crafted creative prompts probe model weaknesses and benchmark defenses	Variable ASR; essential for human-in-the-loop benchmarking
Scenario Nesting (ReNeLLM) [27, 48]	Prompt nesting	ChatGPT, GPT-4, Llama2	●	Black-box	N	LLM generates nested scenarios/roleplay chains to bypass filters	High ASR, fast, generalizes well across contexts

Table 3. Jailbreak attack techniques (part 2), continued from Table 2.

Approach	Attack Type	Target (LLM/Version)	Auto	Threat Model	Grad?	Core Idea / Technique	Outcome
FuzzLLM [181]	Fuzzing	Multiple LLMs	●	Black-box	N	Automated fuzzing with templates, constraints, mutation rules	Efficient discovery with reduced human effort
Atlas (T2I Jailbreak) [29]	Multi-agent/prompt	Text-to-image (T2I) models	●	Black-box	N	LLM-based multi-agent pipeline mutates and filters prompts to bypass T2I safety	Outperforms prior T2I jailbreak approaches
LIAR (Best-of-N Alignment) [5]	Alignment/best-of-N	Multiple LLMs	●	Black-box	N	Efficient best-of-N inference (no training) using diverse prompt candidates	Competitive ASR with ~10× speedup
AutoBreach [12]	Wordplay/mapping	Claude-3, GPT-3.5/4 Turbo, Bing Chat	●	Black-box	N	Wordplay-guided mapping + summarization + two-stage optimization	>80% ASR; <10 queries
GCG (Universal Suffixes) [203]	Suffix optimization	Vicuna, Llama, GPT-3.5, GPT-4, PaLM-2	●	White-box	Y	Greedy coordinate gradient optimization for universal suffixes	87.9% gpt-3.5-turbo-0301; 53.6% gpt-4-0314; 66% PaLM-2 (early-2023 APIs)
AmpleGCG [79]	Suffix generation	Llama-2-7B-chat, GPT-3.5, GPT-4	●	White/Gr	Y	Learns suffix distribution; generates 200 suffixes in 4s (end-to-end: ~6min for 99% ASR)	99% GPT-3.5; ~100% Llama-2
Multi-turn Decomposition [141]	Agent-driven	GPT-3.5-Turbo, GPT-4, Claude	●	Black-box	N	Three-agent system decomposes harmful query into ~10 safe sub-questions; Mixtral-8x7B judge	95.38% GPT-3.5-Turbo; 94.62% Mistral-7B; 93.85% Gemma-2-9B
Jigsaw Puzzles [172]	Multi-turn fragmentation	Multiple LLMs	●	Black-box	N	Splits harmful questions into harmless fragments across conversation turns	Effectively bypasses single-turn safety filters
Multi-Modal Linkage (MML) [158]	Cross-modal	GPT-4o (VLM)	●	Black-box	N	Cross-modal encryption-decryption framing	99.40% SafeBench; 98.81% MM-SafeBench; 99.07% HADES-Dataset
IDEATOR [152]	Visual-language	GPT-4o, Claude-3.5, LLaVA, InstructBLIP	●	Black-box	N	VLM as automated red-team agent generating iterative image-text pairs	94% ASR; 5.34 queries avg.; 82% LLaVA, 88% InstructBLIP transfer
Compositional Instruction [54]	Prompt composition	Multiple LLMs	◐	Black-box	N	Hides harmful intent within complex composite instructions (T-CIA, W-CIA methods)	95%+ on safety datasets; 83%+ GPT-4, 91%+ ChatGPT
Coercive Knowledge Extraction (LINT) [196]	Logit exploitation	Production LLMs	●	Gray-box	N	Exploits output logits to extract harmful knowledge by forcing lower-ranked tokens	92% ASR single attempt; 98% with 5 attempts
Persuasion Attacks [187]	Social engineering	GPT-3.5, GPT-4, Llama-2-7B-Chat	◐	Black-box	N	Leverages 40 persuasion techniques (10 trials/behavior; GPT-4 judge; 1-5 Likert, only 5=jailbreak)	~92% aggregate; GPT-4: 88%, GPT-3.5: 86%, Llama-2: 68%; Claude-1/2: 0%

Concrete mechanisms include JailGuard [190] (universal detection framework), StruQ [14] (Structured Queries defending prompt injection via enforced structured inputs), and JATMO [113] (task-specific finetuning). Effective hardening combines multiple mechanisms rather than single-point solutions; deception-based approaches can systematically overcome isolated defenses. Prompt injection attacks override defensive instructions through crafted inputs [136].

4.1.3 Prompt Adversarial Tuning (PAT). **Prompt Adversarial Tuning** enhances robustness through systematic adversarial example exposure during training/fine-tuning, enabling improved resistance without architectural modifications. PAT demonstrates significant defensive improvements while maintaining standard benchmark performance, indicating adversarial training principles successfully adapt to safety alignment contexts.

Effectiveness depends critically on training example diversity and sophistication, where comprehensive attack pattern coverage enables better novel threat generalization. However, adversarial tuning faces computational costs generating diverse examples and overfitting risks to specific

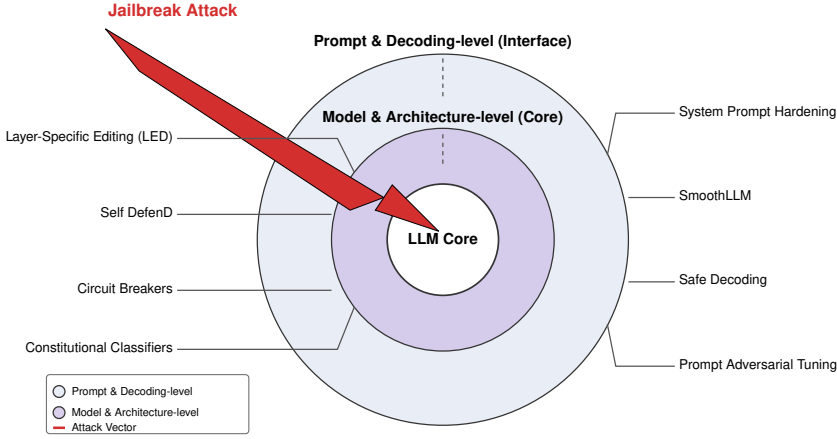


Fig. 8. Defense-in-depth architecture: prompt/decoding-level defenses at interface, model/architecture-level at core.

patterns while remaining vulnerable to novel approaches. Recent research extends adversarial tuning through continuous learning adapting to emerging threats [18].

4.2 Model- and Architecture-level Defenses

Model-level strategies—inner layer in Figure 8—implement security within core architecture and processing pipelines, providing fundamental protection. These prove particularly effective against mechanistic and decoding-time attacks.

4.2.1 Hidden-state Filters and SelfDefenD. **Hidden-state filtering** monitors and validates internal representations during processing identifying and neutralizing adversarial content before influencing outputs. HSF analyzes intermediate activations and attention patterns detecting anomalous processing indicating potential jailbreaks. Representational contrastive scoring detects unseen LVLM jailbreaks efficiently using safety-critical internal layers [42]. Advanced implementations employ ML classifiers distinguishing normal from adversarial patterns while maintaining minimal computational/quality impact.

SelfDefenD [154] runs parallel "shadow" execution isolating and inspecting risky inputs while leaving normal processing untouched. Head-to-head tests reduce attack success on GPT-3.5 by 8.97–97.26% (avg 65.70%) and GPT-4 by 69.69–100% (avg 88.43%), with small overhead (extra delay ≈ 0.06 s GPT-3.5, ≈ 0.35 s GPT-4 under attack; $>95\%$ benign queries incur zero delay). Self-examination where LLMs screen generated responses without fine-tuning demonstrates remarkable effectiveness, reducing attack success to virtually zero (0.02) on GPT-3.5 and Llama 2 [112]. Self-guarding frameworks training LLMs to append safety tags ([harmful]/[harmless]) enable embedded checks without external guardrails [159].

4.2.2 Layer-specific Editing (LED). **Layer-specific Editing** performs local representation edits suppressing harmful behaviors while preserving benign capabilities [198]. On strongly aligned Llama-2-7B-Chat, LED reduces average ASR 17.0% $\rightarrow$ 0.3%; on weaker-aligned Mistral-7B-Instruct 91.2% $\rightarrow$ 11.0% across five attacks (GCG, PAIR suffixes, AutoDAN, GPTFUZZER, DeepInception). DeepInception shows 13.1% residual on Mistral [73, 181]. Utility degradation modest: MT-Bench scores 7.26 $\rightarrow$ 7.08 (-2.5%) Mistral; 6.55 $\rightarrow$ 6.48 (-1.1%) Llama2. False refusal minimal (1–2% estimated). Results reported for *single-turn text-only* attacks with GPT-4 judge; LED not systematically evaluated against multi-turn decomposition [125, 141] or multimodal jailbreaks [88, 105, 158].

Layer-wise pruning identifies "*critical safety layers*" concentrated in early stages that, when removed, cause complete alignment loss even for simple harmful queries. LED edits these crucial layers realigning toxic outputs with safe refusals. Identifying critical early-stage and toxic later-stage layers enables targeted interventions redirecting behavior toward safe outputs without full retraining. Simple parameter editing enables detoxification and resistance using only inference-level resources [150].

LED's mechanistic foundation provides advantages: (1) targeting specific systematically-identified layers minimizes unintended side effects; (2) understanding internal mechanisms informs effective strategies; (3) success across architectures suggests critical safety layers represent generalizable phenomenon. Figure 12 shows LED achieves high effectiveness across attack categories within tested threat model scope.

4.2.3 Constitutional Classifiers and Circuit Breakers. **Constitutional classifiers** train safety filters from natural-language constitutions and large-scale red-teaming data [132]. CBRN-focused deployment reduces jailbreak ASR $\approx 86\% \rightarrow 4.4\%$ on held-out chemical-weapon prompts, using LLM-as-judge evaluation and manual review across thousands of production traffic hours (405 participants, 3,000+ hours, up to \$15,000 per finding). Authors report 23.7% inference overhead and 0.38% false refusal increase, acceptable in high-risk domains. Not evaluated against gradient-based jailbreaks [79, 84, 203]; must be retrained for non-CBRN categories. Authors acknowledge: "We have not studied robustness...to adaptive adversarial attacks targeting these classifiers directly."

Circuit-breaker representation control (Representation Rerouting, RR) reroutes hidden activations from harmful directions using trained "*safety vectors*" [202]. On Mistral-7B-Instruct-v2, rerouting reduces harmful rates $76.7\% \rightarrow 9.8\%$; Llama-3-8B-Instruct $38.1\% \rightarrow 3.8\%$, with Cygnet finetune $\rightarrow 0.8\%$ (97.9% reduction). Multimodal experiments show ASR $91.0\% \rightarrow 14.3\%$ on LLaVA-NeXT-Mistral-7B under strong PGD image perturbations ($\epsilon=32/255$, 1000 steps), with only $\approx 0.5\%$ capability loss (MMMU score $43.8\% \rightarrow 43.3\%$). Tool-using agents on BFCL: direct harmful requests $67.0\% \rightarrow 11.0\%$ (83.6% reduction); forced function calls $64.0\% \rightarrow 11.0\%$ (82.8% reduction); BFCL-AST performance 83.0% maintained.

Caveat: Subsequent work shows adaptive gradient-based attacks regain near-100% ASR optimizing directly against rerouting objective [129]. Independent evaluations report higher false-refusal rates (up to $\approx 38\%$ on large suites like or-bench-80k) than $\approx 3\text{--}4\%$ on WildTeaming-style traffic [53]. RR should be treated as one defense-in-depth layer, not standalone. The 14.3% residual multimodal ASR and documented adaptive vulnerabilities indicate incomplete solutions.

Circuit breaker success across diverse threat models—including multimodal attacks and agent-based vulnerabilities—demonstrates representation-level intervention potential. However, even sophisticated defenses face challenges against advanced feedback-based attacks, highlighting attack-defense asymmetry. Figure 9 illustrates this temporal evolution from 2022-2025, showing how

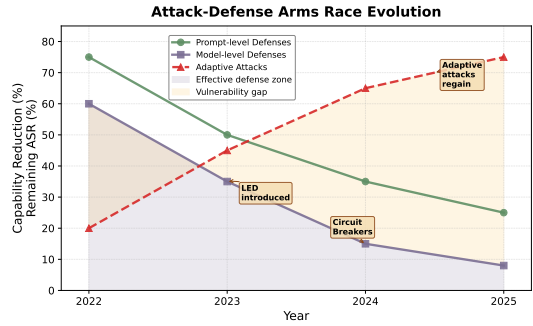


Fig. 9. Temporal evolution of attack-defense dynamics (2022-2025). Prompt-level and model-level defenses progressively reduce ASR, but adaptive attacks exploit mechanistic understanding to regain effectiveness. The persistent vulnerability gap demonstrates the asymmetric advantage attackers maintain despite defensive innovations like LED and Circuit Breakers.

prompt-level and model-level defenses progressively reduce remaining ASR, yet adaptive attacks systematically regain effectiveness.

5 Evaluation Methodologies

5.1 Benchmarks

Systematic evaluation of jailbreak attacks and defensive mechanisms faces challenges that significantly impact the reliability and comparability of research results while constraining meaningful progress in understanding attack effectiveness and defensive improvements [32, 126].

Figure 10 illustrates how judge reliability acts as a foundational dependency that propagates into benchmark validity and metric interpretability. These evaluation challenges encompass benchmark design, judge reliability, metric standardization, and methodological consistency across diverse research contexts, with broader red teaming surveys providing end-to-end perspectives on structuring practical evaluation pipelines [115]. Complementary efforts toward standardized evaluation, such as BackdoorLLM, introduce unified benchmarks that systematize assessment of attack modalities and defenses beyond jailbreak-specific settings [15, 77]. Recent analyses further emphasize that current evaluation frameworks insufficiently capture reasoning-driven vulnerabilities, with reasoning-based backdoors exposing gaps in existing methodological taxonomies [41].

As Figure 11 illustrates, judge reliability issues create a compromised foundation that undermines all three pillars of evaluation: benchmark validity, judge consistency, and metric reliability. This fragility—where automated judge performance varies by over 30 points depending on implementation—constrains research progress by preventing accurate assessment of both attack techniques and defensive strategies.

Contemporary benchmarking efforts attempt to provide standardized evaluation frameworks that enable systematic comparison of attack techniques and defensive mechanisms while ensuring reproducibility and consistency across research efforts. However, existing benchmarks face significant limitations that constrain their effectiveness for comprehensive security assessment, as summarized in Table 4.

5.1.1 JailbreakBench and HarmBench. JailbreakBench represents comprehensive attempts to establish standardized evaluation protocols for jailbreaking research through systematic collection of attack techniques, defensive mechanisms, and evaluation criteria [9]. The benchmark provides curated datasets of harmful behaviors (100 harmful and 100 benign behaviors from JBB-Behaviors), standardized evaluation protocols using Llama-3-70B combined with human annotation, and baseline implementations that enable systematic comparison across different approaches with emphasis on ASR and query cost metrics.

HarmBench introduces standardized evaluation protocols with 510 harmful behaviors across multiple semantic categories while demonstrating that no current attack or defense achieves universal effectiveness across all threat scenarios [95]. The framework provides comprehensive comparison of 18 red teaming methods against 33 LLMs, documenting that robustness remains

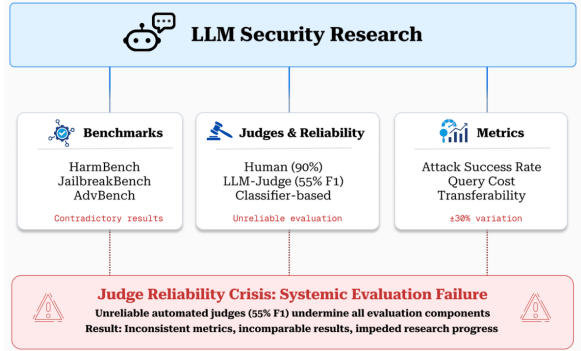


Fig. 10. Judge reliability crisis as compromised foundation undermining evaluation components: benchmarks, judges, and metrics, thereby constraining research progress.

largely independent of model size within families, contradicting assumptions that larger models inherently provide better security. Critical evaluation parameters previously overlooked, such as output token limits, can alter measured ASRs by up to 30%, highlighting how methodological choices impact reported results. HarmBench’s inclusion of novel contextual and multimodal prompts better represents realistic, complex threats compared to traditional text-only benchmarks.

AdvBench provides complementary evaluation through 520 behaviors assessed via string matching and GPT-4 judgment, offering alternative perspectives on assessment methodologies. While simpler than HarmBench, AdvBench’s widespread adoption enables comparison across numerous studies, though with medium reproducibility due to evolving evaluation standards and judge inconsistencies. Additional benchmarking efforts include specialized datasets for cybersecurity contexts [146] and multimodal robustness assessment [88]—the latter proving particularly important for evaluating defenses against cross-modal attacks.

5.1.2 In-the-wild and Specialized Datasets. In-the-wild datasets capture real-world jailbreaking attempts and attack patterns observed in operational deployments, providing crucial insights into practical threat patterns while complementing laboratory-based evaluation approaches [53]. The largest peer-reviewed analysis examined 15,140 prompts across 13 forbidden scenarios, documenting significant discrepancies between academic attack techniques and practical threats while revealing temporal evolution patterns in attack sophistication [135].

Beyond general-purpose benchmarks, specialized resources have emerged [113, 144, 190]. For multimodal robustness, AttackVLM [200] and B-AVIBench [189] offer frameworks for evaluating Large Vision-Language Models against adversarial visual-instructions and black-box attacks. New text-based benchmarks like SafetyBench [195] focus explicitly on evaluating the overall safety of LLMs. Other evaluation suites address specific failure modes, such as XSTest [124] for identifying exaggerated safety behaviors (over-refusal) or the “Are You Human?” [38] adversarial benchmark for exposing LLM weaknesses. The Do-Not-Answer [155] protocol has been introduced as an evaluation method to test LLM safeguards. OpenEval provides comprehensive benchmarking of Chinese LLMs across capability, alignment, and safety dimensions [81].

5.2 Judges and Reliability

The reliability of evaluation mechanisms represents perhaps the most critical challenge facing jailbreaking research, where inconsistent assessment methodologies undermine the validity of reported results while preventing meaningful comparison between different approaches and studies.

5.2.1 Human Annotation and LLM-as-a-Judge. **Human annotation** provides the most authoritative assessment of jailbreaking success through expert evaluation that can consider nuanced contextual factors, semantic implications, and practical harm potential that automated systems may overlook. However, human annotation faces significant scalability limitations that constrain its applicability for large-scale systematic evaluation while introducing potential inconsistencies related to annotator variability, subjective interpretation differences, and evaluation fatigue effects.

LLM-as-a-judge approaches employ AI systems to automatically assess jailbreaking success through analysis of generated content, offering significant scalability advantages while maintaining reasonable accuracy for many evaluation scenarios. However, automated judge reliability spans a wide range. HarmBench’s fine-tuned Llama-2-13B classifier achieves $\approx 93\%$ agreement with human annotations, with GPT-4 judges close behind at $\approx 88\%$ [95]. Simpler heuristics such as substring matching on JailbreakBench and related suites reach $\approx 70\%$ agreement [9], while JAILJUDGE reports vanilla GPT-4 at roughly 0.55 F1 on its most challenging jailbreak-classification tasks before multi-agent scaffolds lift F1 above 0.9 [13, 82]. This 30–40 point spread underpins the evaluation instability highlighted in Figure 11. These limitations indicate that current automated evaluation focuses

on surface-level patterns rather than genuine threat assessment, leading to overestimation of jailbreaking effectiveness and misallocation of defensive resources.

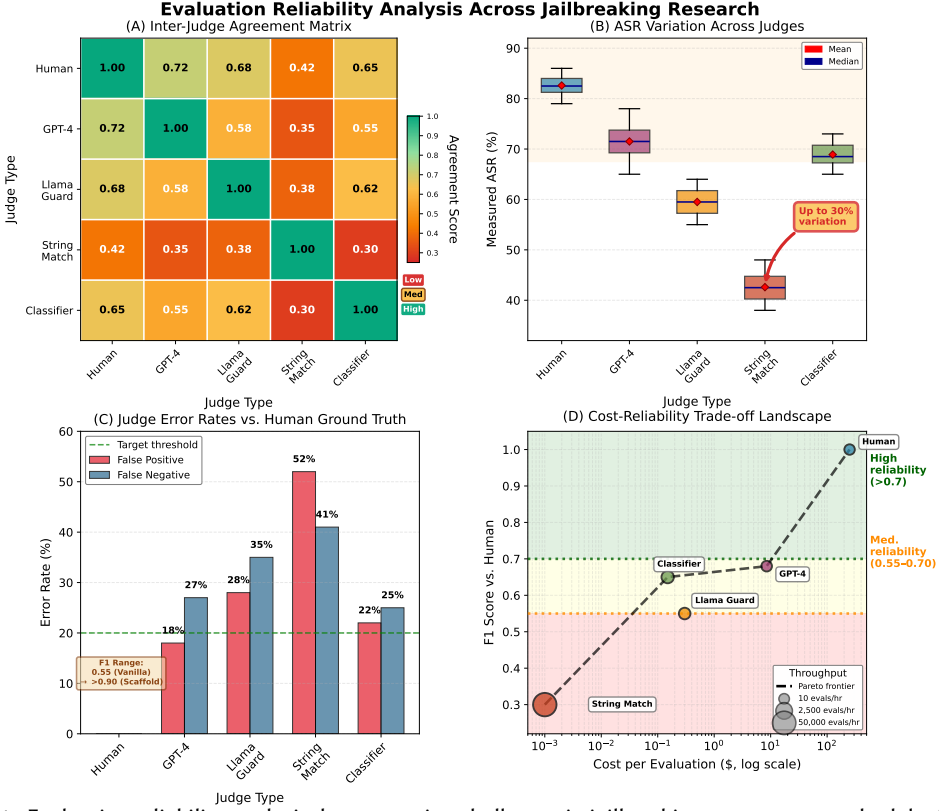


Fig. 11. Evaluation reliability analysis demonstrating challenges in jailbreaking assessment methodologies. (A) Inter-judge agreement matrix shows low concordance (0.30–0.72 F1) between automated evaluation methods and human judgment. (B) Identical attack sets produce up to 30% ASR variation across different judge types, undermining result comparability. (C) Vanilla automated judges exhibit substantial error rates (18–52% false positives, 25–41% false negatives), though performance varies widely (0.55–0.93 F1) with scaffolding. (D) Cost-reliability trade-off landscape demonstrates no universally optimal evaluation approach, with Pareto frontier identifying context-dependent optimal choices ranging from inexpensive string matching (low reliability) to costly human annotation (high reliability).

The reliability challenges illustrated in Figure 11 extend beyond individual judge performance to encompass inconsistent inter-judge agreement and high sensitivity to evaluation methodology choices. Panel A demonstrates that even the most sophisticated automated judges achieve only moderate agreement (0.55–0.72 F1) with human evaluators, while simpler approaches like string matching fall to 0.30–0.42. Panel B quantifies how these differences translate to dramatic ASR variations—up to 30%—across different judge types for identical attack sets. Panel C documents the specific failure modes: automated judges exhibit 18–52% false positive rates and 25–41% false negative rates, indicating they can be fooled by attackers who craft outputs that appear harmful to automated systems while remaining benign to human evaluators. Panel D presents the cost-reliability trade-off, showing that no single evaluation approach dominates across all scenarios.

5.2.2 Calibration Standards. Calibration standards like StrongReject and JAILJUDGE attempt to address evaluation reliability challenges through systematic assessment of judge performance

while providing standardized protocols for evaluation consistency [82]. These frameworks establish baseline performance expectations for different evaluation approaches while offering guidance for selecting appropriate assessment methodologies based on specific research objectives and constraints.

StrongReject introduces stricter criteria for jailbreaking success that penalize superficial violations while focusing evaluation on genuine security concerns that represent practical threats. The framework evaluates 450 samples across 45 categories using GPT-4 combined with human validation, achieving high reproducibility while addressing issues with inflated success rates that result from overly permissive evaluation criteria. JAILJUDGE provides comprehensive benchmarking frameworks for safety judges with human-labeled datasets that document high false positive rates in heuristic approaches and inconsistency in LLM-based evaluation. The emphasis on explainable scoring through multi-agent design enables better understanding of judgment rationales while facilitating debugging and refinement of evaluation procedures. GuidedBench further addresses evaluation discrepancies by equipping assessments with explicit guidelines [46].

5.3 Limitations of Quantitative Synthesis

Our synthesis aggregates results from more than 160 attack and defense papers [23, 30, 58, 169, 177, 178]. We highlight four limitations:

(1) **Heterogeneous conditions.** Studies differ in target models, judge types (human, GPT-4, Llama-Guard, HarmBench classifiers, string matching), sampling parameters, and success rubrics, complicating direct comparison [9, 13, 46, 82, 95, 138].

(2) **Temporal drift.** Commercial APIs evolve; ASRs reported for gpt-3.5-turbo-0301 or gpt-4-0314 may not hold for their 2025 successors.

(3) **Publication bias.** Highly successful attacks and defenses are more likely to be reported [111, 127, 131, 170].

(4) **Reporting gaps.** Many papers omit model versions, decoding settings, or judge prompts, which hinders reproducibility [120, 138].

We therefore report ranges where possible and map each plotted point in Figure 2 to its source, while recommending minimal reporting standards in Table 8.

Table 4. Benchmark and Evaluation Methodology Comparison Across Major Studies

Benchmark/Paper	Harmful Set	Benign Set	Judge(s)	Key Metrics	Reproducibility	Code/Data
JailbreakBench [9]	JBB-Behaviors (100)	JBB-Behaviors (100)	Llama-3-70B + Human	ASR, Query Cost	High	Available
HarmBench [95]	510 behaviors	Multiple contexts	Multiple judges	ASR, Accuracy	High	Available
AdvBench [72]	520 behaviors	Not specified	String matching/GPT-4	ASR	Medium	Available
JailbreakRadar [19]	160 questions (16 categories)	Not specified	GPT-4 Turbo	ASR	Medium	Partial
Crescendo [125]	AdvBench subset	Not specified	Two-stage GPT-4 + Manual	ASR (avg & binary)	High	Available
Safety Misalignment	HarmBench + Custom	Not specified	HarmBench-Llama-2-13b-clb + Human	ASR, Accuracy	Medium	Unclear
JailbreakEval [120]	Toolkit supports multiple	Toolkit supports multiple	Multiple: String, Classifier, LLM	ASR discrepancies	High	Available
In-the-Wild [?]	15,140 prompts (13 scenarios)	Not specified	Internal judge	ASR/ASR-B/ASR-Max	Low	Available
MASTERKEY [25]	Custom dataset	Not specified	Query/Prompt success rates	Query success rates, Prompt ASR	Low	Partial
StrongReject [140]	450 samples (45 categories)	Not specified	GPT-4 + Human	ASR	High	Available
VJailbreakBench [152]	3,654 samples (multimodal)	Not specified	Custom evaluation	ASR	Medium	Unclear
SafeBench [158]	Multimodal safety	Multimodal benign	GPT-4 judge	ASR	Medium	Unclear
Wildteaming [53]	In-the-wild jailbreaks	Not specified	Multiple approaches	ASR, Safety improvement	High	Available
LatentJailbreak [118]	Text safety behaviors	Multiple contexts	Automated evaluation	Robustness metrics	Medium	Available
OpenEval [81]	Chinese LLM behaviors	Capability + alignment	Multi-dimensional	Safety, Alignment, Capability	High	Available
DecodingTrust [147]	Multiple perspectives	Multiple contexts	Comprehensive framework	Trustworthiness across 8 dimensions	High	Available

6 Quantitative Trends and Comparative Analysis

Systematic analysis of quantitative trends demonstrates consistent evolutionary dynamics in attack sophistication, defensive development, and evaluation methodology [69]. This synthesis integrates temporal trends with cross-technique comparisons, quantifying effectiveness patterns across the threat landscape.

6.1 Attack Evolution and Success Patterns

Contemporary attacks demonstrate improving effectiveness with predictable scaling. Figure 2 shows recent proposals report very high ASRs on specific model-judge-benchmark combinations: AmpleGCG reaches $\approx 99\%$ on Llama-2-7B-Chat (200–1000 suffixes, AdvBench) [79]; TAP attains 90–94% on GPT-4/4-Turbo/4o under multi-turn protocols [98]; multimodal linkage exceeds 94–99% on VLM benchmarks [152, 158]. Multi-agent debate architectures show amplified vulnerabilities, with structured attacks raising harmfulness from $\approx 28\%$ to $>80\%$ [116].

Threat-model stratification. Figure 2 aggregates attacks under different access assumptions. White-box gradient attacks (GCG, mechanistic interventions) achieve $\approx 90\text{--}99\%$ ASR on open-weight models [162, 203], serving as upper-bound probes. Gray-box attacks exploit log-probabilities or prefill [57, 76, 90]. Black-box query-only attacks (PAIR, TAP, fuzzing, in-the-wild prompts) [10, 35, 70, 98, 135] climbed from $\approx 40\%$ to 80–94% on modern GPT-4-class APIs. Black-box trends are operationally most relevant for commercial systems; white-box results inform training-time robustness.

Multimodal attacks demonstrate 94–99% effectiveness against VLMs, confirming cross-modality alignment challenges. Multi-Modal Linkage achieves near-perfect rates via encoding-decoding exploitation. Results on commercial models highlight multimodal safety remains unsolved despite extensive investments.

Automated generation shows remarkable efficiency improvements. AmpleGCG samples ~ 200 suffixes in 4 s inference (end-to-end: several minutes for $\approx 99\%$ ASR batch) [79]; JBFuzz achieves 99% average ASR with ~ 60 s per jailbreak [35]. Automation enables rapid vulnerability discovery without manual optimization. PAIR demonstrates \$0.026 per attack (<20 queries, Mixtral-8 $\times$ 7B $\rightarrow$ Vicuna-13B-v1.5) [10], eliminating economic barriers.

6.2 Transferability Patterns

Transferability depends on architectural similarities, training, and defenses. Pretrained weight exposure can yield jailbreak prompts that transfer strongly to finetuned variants [142]. GCG-style suffixes achieve 87.9% ASR on gpt-3.5-turbo-0301, 53.6% on gpt-4-0314, 66% on PaLM-2 with ensembles [203]—though early-2023 snapshots may overestimate current performance. White-box gradient approaches show improved transferability when superfluous constraints are removed (18.4% $\rightarrow$ 50.3% target ASR) [173].

Multimodal attacks demonstrate strong transferability. IDEATOR prompts achieve 82% ASR on LLaVA, 88% on InstructBLIP when trained on other VLMs [152], indicating cross-modality inconsistencies represent systematic vulnerabilities rather than implementation artifacts.

6.3 Defense Progress and Limitations

Defensive mechanisms show progress balancing security with performance. Layer-specific Editing (LED) reduces average ASR from 17.0% to 0.3% on Llama-2-7B-Chat and from 91.2% to 11.0% on Mistral-7B-Instruct across five attacks, though DeepInception maintains 13.1% residual on Mistral [73, 181, 198]. Reasoning-enhanced fine-tuning (RATIONAL) improves robustness and helpfulness-safety trade-offs via explicit safety reasoning [193]. SelfDefenD achieves 65–88% reduction with minimal latency. Circuit breakers provide orders-of-magnitude improvement across single-turn, multi-turn, multimodal, and automated attacks.

Persistent challenges remain. Feedback-based attacks (PAIR, TAP) retain residual ASR often $>15\%$ under combined system prompts, classifiers, and safe decoding [19], demonstrating sustained effectiveness against layered mechanisms. Generation-parameter attacks achieve 68% average

ASR, highlighting inadequately addressed vulnerabilities. Compression may reduce toxicity but exacerbate bias [167]. This asymmetry necessitates security-by-design architectures.

6.4 Defense Effectiveness Across Attack Categories

Figure 12 summarizes relative ASR reduction across attack categories. Mechanistic defenses (LED, Circuit Breakers) generally outperform prompt-level defenses against specific single-turn attacks (GCG, AutoDAN), often reducing ASR to $<5\%$ in white-box settings [198, 202].

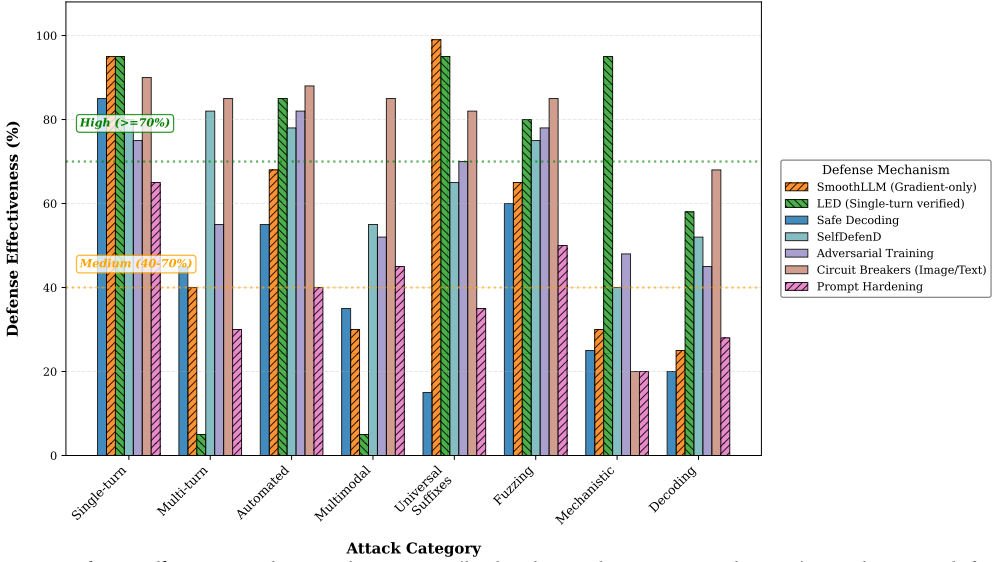


Fig. 12. Defense effectiveness by attack category (higher bars = larger ASR reduction). Mechanistic defenses outperform prompt-level across most types; multi-turn and multimodal attacks remain resilient. Values aggregated from reported results, sensitive to judge choice, refusal rubric, decoding settings.

Gains are threat-model specific. Defense effectiveness varies by attack sophistication: LED maintains 13.1% ASR on DeepInception (Mistral); Circuit Breakers show 9.8–14.3% residual across text/multimodal but face 100% ASR under adaptive attacks [129]; SmoothLLM drops GCG to $<1\%$ but PAIR maintains 56% ASR; SafeDecoding achieves 0% on DeepInception but experiences 12.4% semantic transition failures. Defenses optimized for gradient attacks often fail against semantic persuasion.

Single-turn attacks are most amenable to mitigation; multi-turn, automated, multimodal remain challenging. SmoothLLM reports 100× reductions on Llama 2 and 50× on Vicuna against GCG, illustrating character-level smoothing disrupting token-level optimization. Headline reductions are setup-dependent: measured ASR varies with judge, rubric, and parameters. Figure 12 should be interpreted directionally.

No single mechanism provides universal protection, necessitating layered strategies. Heuristic attacks achieve exceptional rates but remain fragile; feedback attacks generate diverse prompts maintaining effectiveness under layered protections [19]. Mechanistic approaches (LED, circuit breakers) require calibration balancing security, performance, and operational feasibility. Table 5 summarizes representative jailbreak attack methods, their threat models, and practical trade-offs (ASR, budget, transferability, detectability).

Complementing the attack landscape in Table 5, Table 6 compares major defenses by effectiveness across attack types, tested threat models, and known gaps.

Table 5. Comparative Analysis of State-of-the-Art Jailbreak Attack Methods

Attack Method	Category	Threat Model	ASR (Dataset/Model/Judge)	Query Budget	Transferability	Detectability	Complexity	Code/Data
GCG [203]	Prompt-based	White-box (Grad)	87.9% GPT-3.5, 53.6% GPT-4, 66% PaLM-2	Medium	High	High	Low	Available
AmpleGCG [79]	Automated	White/Gray	99% GPT-3.5, ~100% Llama-2	4 sec/200 suffixes	Very High	High	Medium	Available
Multi-turn Decomp. [141]	Agent-based	Black-box	95.38% GPT-3.5-Turbo, >90% typical	High	Medium	Low	Medium	Available
TAP [98]	Tree Search	Black-box	>80% (GPT-4o/GPT-4 judge)	60% fewer vs. PAIR	High	Medium	High	Partial
PAIR [10]	Query-efficient	Black-box	88% Vicuna, ~50% GPT-3.5/4, 73% Gemini	<20 queries, \$0.026	Medium	Medium	Medium	Available
MASTERKEY [25]	Automated	Black-box	21.58% avg., 14.51% Bard, 13.63% Bing	Medium	Medium	Low	High	Partial
MML [158]	Multimodal	Black-box	99.4% SafeBench, 98.81% MM-SafeBench	Medium	Medium	Low	High	Unclear
IDEATOR [152]	Multimodal	Black-box	94% (5.34 queries), 46.31% GPT-4o	Low	High	Low	High	Unclear
JBfuzz [35]	Fuzzing	Black-box	Avg. 99% across 9 LLMs	~60 s per jailbreak	High	Medium	Medium	Available
Jigsaw Puzzles [172]	Multi-turn	Black-box	High across multiple LLMs	Medium	High	Low	Medium	Partial

Table 6. Defense Mechanisms Comparison: Effectiveness, Tested Threat Models, and Known Gaps

Defense Method	Single-Turn	Multi-Turn	Multimodal	Tested Threat Models	Untested / Gaps	Overhead	Deployability
Safe Decoding [169]	High (96%)	Medium	Low	White-box (Logits); Single-turn suffix	Semantic transition failures (~12%); Closed APIs	Low	Low (Logits req)
SmoothLLM [123]	High (99%)	Medium	Low	Black-box query; Gradient-based suffix	Semantic persuasion (PAIR); High latency cost	High (N×)	High
System Prompting [192]	Medium	Low	Medium	Black-box query; Instruction following	Agentic decomposition; Encoding attacks	Minimal	Very High
LED [198]	High (0.3%)	Untested	Untested	White-box (Weights); Single-turn text	Multi-turn context; Multimodal inputs; Agents	Low	High (Open wts)
SelfDefenD [154]	High (65–88%)	High	Medium	Black-box; Shadow stack	High-throughput latency; Resource exhaustion	Medium	High
Circuit Breakers [202]	High (9.8%)	Medium	High (14.3%)	White-box (Rep. Eng.); Text/Image/Agent	Adaptive gradient attacks (100% ASR)	Low	High
Self-Examination [112]	Very High	High	Medium	Black-box; Response filtering	False refusal rates; Inference cost	Medium	High
Model Surgery [150]	High	Medium	Medium	White-box (Params); Toxicity	Complex logical jailbreaks	Low	High

Table 7 summarizes attack/defense success rates and deployment trade-offs.

7 Limitations and Open Challenges

Jailbreaking research faces persistent limitations constraining progress and highlighting areas requiring innovative approaches. Figure 13 taxonomizes interconnected challenges spanning evaluation, attack evolution, and defense domains.

7.1 Defense Effectiveness and Limitations

Comprehensive evaluations document that no single mechanism provides universal protection against all attack types. Systematic frameworks assess defenses across safety and over-defensiveness risk [144]. Layered strategies combining complementary approaches prove most effective, yet face persistent challenges against sophisticated attacks—particularly feedback-based approaches (PAIR, TAP) maintaining effectiveness through natural, diverse prompts.

These observations motivate security-by-design architectures, integrating defensive mechanisms throughout model development rather than as external patches.

7.2 Judge Fragility and Standardization

Judge reliability represents the most critical limitation constraining systematic progress. Automated judge reliability spans wide ranges: HarmBench’s fine-tuned Llama-2-13B classifier achieves ~93% agreement with human annotations; GPT-4 judges ~88% [95]. Substring matching reaches ~70% agreement [9]; vanilla GPT-4 shows roughly 0.55 F1 on challenging tasks before multi-agent scaffolds lift F1 above 0.9 [13, 82]. This 30–40 point spread underpins evaluation instability in

Table 7. Comparison of Attack and Defense Methods: Effectiveness, Success Rates, and Trade-offs

Method/Approach	Attack Success Rate	Defense Success Rate	Notable Trade-offs/Limitations	Citations
Gradient-based Attack	High (87.9% GPT-3.5, 53.6% GPT-4)	Low	Requires model access for white-box; high transferability	[203] [19]
Heuristic/Template	Very High (100% DeepSeek-V3)	Low (fragile)	Easy to mitigate, limited diversity makes defenses effective	[19] [58]
Multi-turn/Agent-based	Very High (95.38% GPT-3.5)	Medium (ASR>15% layered)	Effective against layered defenses; requires context tracking	[141] [19] [172]
Multimodal Attack	Exceptional (99.4% SafeBench)	Low	Exploits cross-modal gaps; current defenses inadequate	[158] [152]
Prompt Filtering	N/A	Moderate	May reduce utility; vulnerable to encoding attacks	[111] [131] [192]
Model Fine-tuning	N/A	Best overall (near-zero with LED)	May degrade performance; LED minimizes impact	[198] [92] [111] [58]
Intention Analysis	N/A	High (~48.2% ASR)	Inference-only, robust to errors; computationally efficient	[192] [111]
Response Evaluation	N/A	Moderate-High	Latency/resource overhead; judge reliability concerns	[131] [19] [58]
Semantic Augmentation	N/A	Medium-High	Improves harmful content detection via LLM-based data cleaning	[97]
Alignment-Aware Quantization	N/A	High	Preserves safety during compression; prevents degradation	[160]
Pretraining Filtering	N/A	High (preventive)	Addresses root causes; requires webscale dataset curation	[99]

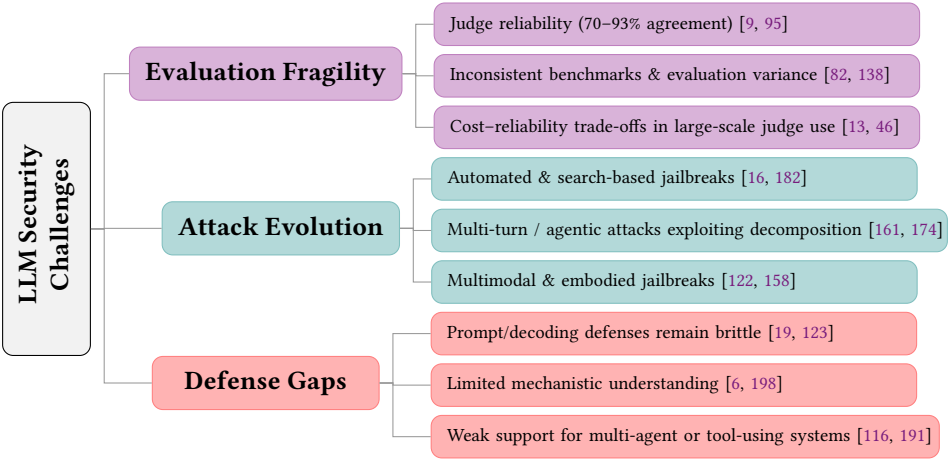


Fig. 13. Hierarchical view of key LLM security challenges: evaluation fragility, attack evolution, and defense gaps.

Figure 11, manifesting as inter-judge disagreement (0.30–0.72 F1), dramatic ASR variations (up to 30%), and substantial error rates (18–52% false positives).

Evaluation fragmentation prevents systematic meta-analysis or cumulative knowledge development. Addressing judge fragility requires coordinated community efforts establishing standardized protocols, comprehensive benchmarks, and calibrated methodologies enabling reliable cross-study comparison [46, 82].

7.3 API Evolution and Reproducibility

Commercial AI system dynamics create reproducibility challenges where techniques effective at specific times become ineffective as systems evolve. API versioning, defensive updates, and policy modifications dramatically alter behavior, invalidating previous results and complicating longitudinal analysis. This temporal instability explains effectiveness variation across different time points.

Reproducibility challenges encompass access limitations, rate limiting, cost constraints, and ethical considerations preventing comprehensive evaluations. Establishing reproducible practices requires coordination between researchers, providers, and standardization organizations.

7.4 Multi-turn and Agentic Risk

Current approaches focus on single-turn interactions while inadequately addressing multi-turn conversational and agent-based vulnerabilities [26, 71, 86, 183]. Agent-driven decomposition achieving 95.38% ASR demonstrates temporal attack distribution bypasses defenses designed for isolated queries [141]. Jigsaw Puzzles splitting queries into fragments illustrates single-turn filter limitations [172]. Agent-based vulnerabilities integrating external tools create expanded attack surfaces including Action Hijacking [191], multimodal agent attacks [66, 163], and Foot-in-the-Door sequential reasoning exploits compromising ReAct agents [104].

RoboPAIR demonstrates jailbreaks translating to harmful physical robotic actions, achieving up to 100% ASR [122]. Emerging threats target RAG and memory components [17, 52], including RAG-Thief data extraction, information flooding in multi-agent communities [61], and memory poisoning. Addressing multi-turn and agentic risks requires defensive architectures maintaining comprehensive conversational context analysis with robust containment—capabilities current defenses lack.

7.5 Multimodal Safety Gaps

Multiple modality integration challenges consistent safety across processing pathways. Current alignment focuses on individual modalities while inadequately addressing cross-modal attacks achieving near-perfect rates via encoding-decoding exploitation. Strong transferability—82% ASR on LLaVA, 88% on InstructBLIP for IDEATOR prompts—indicates cross-modal vulnerabilities represent alignment challenges rather than model-specific weaknesses.

Figure 2 shows multimodal attacks achieving 99% effectiveness by 2025; Figure 12 shows current defenses achieve only low-to-medium effectiveness. Guardrail impacts reveal tensions between safety and effectiveness in domains like counterspeech [7], with bias concerns in transgender/non-binary representation [107]. Controllable toxicity avoidance preserving performance offers promising directions [64].

7.6 Mechanistic Understanding vs. Ad-hoc Patching

Current research trends toward ad-hoc solutions addressing specific attacks without understanding underlying mechanisms. This reactive approach creates attack-defense cycles without systematic progress toward robust architectures. Feedback-based attacks retaining >15% effectiveness against layered defenses while generation-parameter attacks achieve 68% average ASR highlights how prompt-level defenses fail [19].

Layer-specific Editing demonstrates mechanistic understanding advantages by identifying critical safety layers through systematic analysis, achieving near-zero ASR with minimal helpfulness impact [198]. This contrasts with blanket modifications lacking mechanism comprehension. Mechanistic understanding requires research into computational and architectural factors enabling vulnerabilities while developing theoretical frameworks guiding systematic security enhancement [6]. Understanding internal mechanisms—particularly how hidden states reveal ethical concepts learned during pre-training with alignment associating these to emotion guesses [201]—can inform both attack and defense strategies.

8 Future Directions

Critical limitations in jailbreaking research establish priorities for advancing LLM security in high-stakes applications. Building on attacks, defenses, evaluation challenges, trends, and limitations, we establish research priorities for secure AI development.

8.1 Toward Security-by-Design in LLMs

Security-by-design shifts from reactive mechanisms toward proactive architectural solutions integrating security throughout model design, training, and deployment [151]. LED’s mechanistic approach demonstrates how understanding internal mechanisms informs targeted interventions achieving near-zero ASR while maintaining helpfulness [198].

Future research must embed security mechanisms at fundamental levels rather than as external add-ons. This includes training methodologies inherently producing robust models [157], architectural modifications limiting attack surfaces, and optimization balancing capability with security. Constitutional AI with formal verification offers promising directions. Certifiable safety guarantees against adversarial prompting through token erasure represent concrete verification steps [65]. Filtering harmful content from webscale pretraining datasets addresses root-level safety [99], while alignment-aware quantization preserves safety during compression [160]. LLM salting aims to reduce jailbreak reuse and transferability by introducing deployment-specific randomness into refusal behaviors, complicating precomputed jailbreak portability [145].

8.2 Unified Benchmarks with Human-Calibrated Judges

Unified evaluation frameworks represent critical priority for systematic progress, ensuring reported results provide meaningful security insights. Benchmarking must integrate comprehensive attack coverage with reliable assessment combining automated efficiency and human accuracy [9, 95]. Evaluation challenges—particularly judge reliability crisis (Figure 11)—necessitate immediate community action.

Table 8 outlines standardized reporting requirements enabling meaningful cross-study comparison while ensuring reproducibility and temporal validity.

Table 8. Minimum Required Reporting Standards for Jailbreaking Research

Metric Category	Required Details
Threat Model & Access	Specify: Black-box (query-only), Gray-box (logprobs/prefill), or White-box (gradients/weights/hidden states). State whether results apply to open-weight models, closed APIs, or both.
Cost Metrics	Query budget, token budget, computational latency (per attack/query), total resource requirements (e.g., GPU hours for generation).
Judge Configuration	Judge type (human/LLM/classifier), specific model version (e.g., GPT-4-0613), exact system prompt, success threshold definition, and inter-annotator agreement stats.
System Context	Target model version/timestamp (e.g., gpt-3.5-turbo-0125), API endpoint, decoding parameters (temp, top-p), and defensive mechanisms active.
Evaluation Setup	Dataset specification with version (e.g., AdvBench-50 vs. full), evaluation protocol details, and statistical significance testing.
Reproducibility	Code availability, data accessibility, experimental configuration files, random seeds, and environment specifications.

8.3 Formalizing Threat Models for Multi-agent and Tool Use

Agent-based vulnerabilities and multi-turn attacks necessitate threat modeling encompassing expanded attack surfaces from AI integration with external tools, APIs, and execution environments.

Standardized tool mediation layers (e.g., **Model Context Protocol**, MCP) shift prompt injection into a protocol layer, where tool descriptions, schemas, and tool outputs can carry adversarial instructions (“*tool poisoning*”) and induce unauthorized actions [180]. Recent evidence suggests such protocol-layer injections can target *tool invocation decisions* (not only output text), enabling covert exfiltration through logging-style tools while preserving apparent task quality. Multi-agent deployments further expand the attack surface: **agent-to-agent delegation** can amplify privilege mismatches, where a less-privileged agent is induced to recruit a more-privileged agent to perform restricted actions, and inter-agent communication channels themselves become manipulable objects of attack [11]. These agentic and tool-mediated vectors align with operational risk taxonomies that treat prompt injection as a top-tier deployment risk [108].

Research must develop formal frameworks analyzing agent-based vulnerabilities while establishing security protocols preventing unauthorized access or harmful actions. This includes containment mechanisms isolating harmful operations, privilege management enforcing least-privilege principles, and monitoring frameworks for AI agent deployments. For this purpose, the formal model introduced in [68] to describe the arms race in the context of adversarial malware detection can serve as a starting point.

Multi-agent threat modeling must account for compositional vulnerabilities where individually safe components produce dangerous outcomes. Three-agent decomposition fragmenting harmful queries into safe sub-questions demonstrates compositional attacks exploiting limited holistic reasoning [141]. Jigsaw Puzzles fragmentation further illustrates compositional risks [172]. Defensive frameworks must analyze multi-component interactions while maintaining real-time computational efficiency.

8.4 Mechanistic Defenses with Utility Guarantees

Mechanistic defense development requires understanding computational and architectural factors enabling vulnerabilities while developing principled approaches providing theoretical guarantees. Research must transition from empirical development toward theoretically grounded approaches providing formal security guarantees while maintaining model utility. This includes verification methodologies proving defensive effectiveness under specified threat models [101], formal analysis identifying vulnerability sources through architectural inspection, and architectural principles enabling systematic security enhancement.

Mechanistic interpretability success enabling attacks achieving 80–95% ASR within minutes [162] and defenses achieving near-zero ASR with minimal utility impact [198] suggests understanding internal mechanisms represents crucial research direction. Intermediate hidden state analysis revealing ethical concepts learned during pre-training and associated with emotion guesses during alignment [201] provides concrete mechanistic foundations. Complementarily, self-aware guard enhancement shows that models may detect unsafe intent yet fail to act on that knowledge, motivating mechanisms that convert internal risk awareness into reliable refusal behavior [28].

8.5 Integration into Secure Systems Frameworks

LLM security must integrate within broader cybersecurity frameworks addressing interactions between AI systems and surrounding infrastructure, applications, and operational contexts [102]. For tool-using assistants, especially MCP-like ecosystems, secure deployment requires tool authentication, capability-scoped authorization, provenance-aware mediation, and auditable logging of tool calls and tool outputs. For multi-agent systems, these controls must extend to **agent identities** and **agent-to-agent messages**, including integrity-protected tool schemas, supervised execution for privileged actions, and explicit delegation policies that prevent lower-trust agents from escalating via higher-privilege agents.

Time-based analysis reverse-engineering defensive mechanisms highlights AI security intersecting traditional cybersecurity concerns like information leakage through side channels [45]. Programmatic behavior exploitation demonstrates instruction-following amplifying standard security risks [62].

Research must develop comprehensive security architectures encompassing AI-specific threats while integrating traditional cybersecurity approaches, providing defense-in-depth strategies addressing full attack vector spectra [2, 63]. This includes security monitoring detecting anomalous AI behavior [139], incident response protocols for AI security breaches, and security testing methodologies systematically assessing vulnerabilities before deployment.

9 Conclusion

This survey documents a pronounced asymmetry in the LLM security landscape. Automated black-box attacks now report success rates in the 80–94% range on deployed APIs, while white-box techniques reach $\approx 99\%$ on open-weight models. Multimodal jailbreaks achieve up to 99.4% success in specific benchmarks, and agent-driven decomposition exploits context fragmentation to bypass filters with $>95\%$ effectiveness. Simultaneously, the field faces a reliability crisis in evaluation: automated judge agreement with human labels varies from 70% to 93% depending on implementation, introducing significant variance in reported results.

Robust LLM security demands a paradigm shift toward architectures built on three pillars: (i) unified, stratified evaluation that clearly distinguishes between theoretical white-box vulnerabilities and practical black-box risks; (ii) mechanistic defenses (e.g., LED, Circuit Breakers) that target internal representations rather than surface-level prompts, while acknowledging their current limitation to specific threat models; and (iii) comprehensive threat modeling that explicitly addresses multi-turn, agentic, and multimodal vectors. Only by moving from ad-hoc patches to security-by-design principles can we ensure the safe deployment of LLMs in high-stakes environments.

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